FINAL PROJECT REPORT

Global Fuel Prices and Taxation Trends: A Comprehensive Global Performance Intelligence Report

Submitted by
Harshvardhan Gupta
Bachelor of Computer Application (BCA)
3rd Semester | MEDI-CAPS University

Project Type: Individual
Platform: Tableau Public 2026.3
Data Source: global_fuel_prices_Dataset.csv

Purpose: To learn the concepts of Tableau and data analytics through exploratory analysis of global fuel prices, taxation, subsidies, and economic patterns.

Report Structure

- 1. Introduction
- 2. Ideation Phase
- 3. Requirement Analysis
- 4. Project Design
- 5. Project Planning & Scheduling
- 6. Functional and Performance Testing
- 7. Results
- 8. Advantages & Disadvantages
- 9. Conclusion
- 10. Future Scope
- 11. Appendix

Executive Summary

This individual project uses Tableau Public 2026.3 to explore a supplied historical dataset of global fuel prices and taxation. The CSV includes country, region, income level, subsidy level, petrol, diesel and LPG prices, Brent crude price, tax percentage, and date. The workbook contains three scenario-focused dashboards and a supporting historical trend worksheet. The dashboards organize country and regional comparisons, tax and subsidy exploration, fuel-price rankings, and income-level comparisons in a published Tableau Public workbook.

The project is an exploratory business-intelligence report rather than a live monitoring system or causal economic study. The dataset is connected directly from CSV; a separate database, automated external feed, custom web application, and formal load-testing results are not part of the confirmed implementation.

1. INTRODUCTION

1.1 Project Overview

Fuel prices are influenced by multiple factors, including international crude oil prices, taxes, subsidies, economic conditions, and geographic differences. Comparing these indicators across many countries and over time can be difficult when the information is viewed only as a raw table. This project converts the supplied dataset into interactive Tableau dashboards to make comparisons easier to explore.

The dataset notes describe approximately 27,468 records across 84 countries and 7 regions, covering January 2020 to April 2026. Fields include date, country, region, income_level, subsidy_level, petrol_usd_liter, diesel_usd_liter, lpg_usd_liter, brent_crude_usd, and tax_percentage.

1.2 Purpose

The primary purpose is to develop practical knowledge of Tableau and data analytics by connecting a structured CSV dataset, creating visualisations, arranging dashboards, applying available filters and legends, and publishing the workbook on Tableau Public.

- Compare fuel-tax percentages and petrol prices across countries and regions.
- Explore high- and low-price country groups and compare petrol and diesel prices. Examine fuel prices by subsidy level and compare patterns across income groups.
- Explore historical fuel-price movements alongside Brent crude prices without treating association as proof of causation.

2. IDEATION PHASE

2.1 Problem Statement

Problem: The dataset contains many countries, dates, regions, subsidy categories, and income groups. Reading these fields in raw tabular form can make comparisons difficult. The project therefore needs a clear visual interface that brings related measures together while preserving careful interpretation of patterns.

Scenario	Intended user need	Dashboard response
1 — Policy	Compare fuel taxation, regional petrol prices, and subsidy categories.	Country tax ranking, regional average petrol prices, tax-versus-price scatter plot, and prices by subsidy level.
2 — Supply chain	Identify relatively high- and low-price countries and explore regional and historical price patterns.	Top and bottom country rankings, regional average petrol prices, and Brent crude/fuel-price trend view.
3 — Economic development	Compare average fuel prices and taxes across income groups and subsidy categories.	Average petrol price by income level, average tax by income level, and subsidy/tax comparisons with filters.

2.2 Empathy Map Canvas

Three project-defined analytical personas guided the dashboard scenarios: an energy policy consultant, a global supply chain manager, and an energy economics researcher. These personas are design aids, not real interview participants or evidence of field research.

- Likely needs: readable comparisons, clear units, meaningful titles, and filters that narrow the analysis.
- Likely pain points: many countries and categories, difficulty spotting patterns in tables, and risk of overinterpreting correlations.
- Design implication: use rankings, grouped comparisons, trend charts, consistent labels, and visible legends where applicable.

2.3 Brainstorming

Idea	Category	Decision
Compare tax percentage with average petrol price.	Taxation and policy	Selected for Scenario 1.
Compare country and regional fuel prices.	Regional comparison and supply chain	Selected for Scenario 2.
Explore fuel prices by subsidy and income group.	Subsidy and economic development	Selected for Scenario 3.
Compare historical petrol/diesel prices with Brent crude.	Time-series analysis	Retained as a supporting trend worksheet/view.

The ideas and priorities were documented individually. They are not represented as the output of a team brainstorming session or measured user research.

3. REQUIREMENT ANALYSIS

3.1 Customer Journey Map

Stage	User activity	Dashboard support
Access and orient	Open the published workbook and choose the relevant scenario.	Three dashboards with descriptive titles.
Explore and compare	Read chart titles, compare categories, and use visible filters or legends.	Bar charts, scatter plot, ranked country views, trend chart, and visible filters.
Interpret and decide	Describe patterns and identify questions for further analysis.	Grouped comparisons and historical trends; interpretation remains descriptive.

This is a proposed journey based on intended use, not a journey validated through user interviews or formal usability testing.

3.2 Solution Requirements

Functional requirements include connecting the supplied CSV, using the available fields in worksheets, comparing average prices and tax percentages, organizing visualisations into three dashboards, applying available filters, and publishing the workbook for viewing. Non-functional considerations include readability, understandable labels, reliability of the source data, appropriate access, and performance. Formal testing has not established response-time or concurrency thresholds.

3.3 Data Flow Diagram

Implemented data flow: supplied CSV file → Tableau Public data connection and field configuration → worksheets and aggregations → three scenario dashboards → published Tableau Public workbook. No separate database, Snowflake environment, SQL pipeline, external ETL tool, or live price API is claimed.



Data flow is a static CSV-to-Tableau workflow; published views are not a real-time feed.

3.4 Technology Stack

Component	Technology / approach	Use in project
Source data	CSV: global_fuel_prices_Dataset.csv	Structured historical data for analysis.
BI and visualisation	Tableau Public 2026.3	Worksheets, charts, dashboard layout, filters/legends, and publication.
Processing	Tableau data-source settings and built-in aggregations	Use field roles/types, average measures, rankings, and dashboard filters as applicable.
Publication	Tableau Public workbook	Public link to the three scenario dashboards.
Not used / not confirmed	Separate database, SQL, Python ETL, live API, custom web app	These are not part of the confirmed current implementation.

4. PROJECT DESIGN

4.1 Problem–Solution Fit

The proposed solution matches the three analytical needs by consolidating relevant measures into visual comparisons. This fit is reasoned from the requirements and implemented dashboard features; it has not been validated through customer interviews, surveys, or a formal usability study.

4.2 Proposed Solution

The solution is a Tableau Public workbook containing three dashboards, each designed around a scenario. A supporting worksheet named "Global Petrol Price Trend" is retained in the workbook but is not counted as a fourth dashboard.

Dashboard	Main visualisations	Interaction / interpretation
Scenario 1 — Fuel Taxation & Subsidy Policy	Top 10 countries by average fuel tax percentage; average petrol price by region; taxation vs petrol price; fuel prices by subsidy level.	Region legend; scatter plot supports exploration of association, not causation.
Scenario 2 — Global Fuel Prices & Supply Chain Planning	Top 10 and bottom 10 countries by average petrol price, with petrol/diesel comparison; average petrol price by region; Brent crude and fuel-price trends.	Region and Measure Names legends; country rankings are based on average petrol price.
Scenario 3 — Fuel Prices, Taxation & Economic Development	Taxation vs petrol price; fuel prices by subsidy level; average petrol price by income level; average tax percentage by income level.	Visible Income Level, Region, and Subsidy Level filters; group averages can hide within-group variation.

4.3 Solution Architecture

The workbook uses a direct file-based connection. The data source feeds Tableau worksheets; worksheets are arranged into dashboards, and the workbook is published through Tableau Public. The architecture is intentionally simple and appropriate to the current project scope. The workbook is public, so confidential or restricted data must not be added to it.

5. PROJECT PLANNING & SCHEDULING

5.1 Project Planning

A planning backlog was organized around data access, worksheet creation, dashboard construction, publication, and documentation. An illustrative four-sprint plan was prepared, with a total of 38 estimated story points. These points are relative planning estimates; formal sprint completion records, actual velocity, and a measured burndown history were not maintained.

Workstream	Planned / completed activity	Evidence / status
Data access	Connect the supplied CSV and review fields.	CSV connected directly to Tableau Public.
Analysis	Create worksheets for tax, price, subsidy, region, income level, and historical comparisons.	Supporting worksheets used in the three dashboards.
Dashboard design	Arrange four charts per scenario and use relevant legends/filters.	Three scenario dashboards documented with current screenshots.
Publication	Publish the workbook and include all relevant worksheets.	Published Tableau Public workbook link provided in Appendix.
Documentation	Prepare planning, design, testing, demonstration, and report documents.	Documentation compiled; GitHub link was not provided.

6. FUNCTIONAL AND PERFORMANCE TESTING

6.1 Performance Testing

Available evidence confirms the dataset connection, dashboard structure, visualisation types, and visible filters/legends in the supplied screenshots. It does not establish measured rendering speed, query latency, concurrency, or mobile responsiveness. Therefore, this report records evidence-based functional observations rather than inventing numerical performance results.

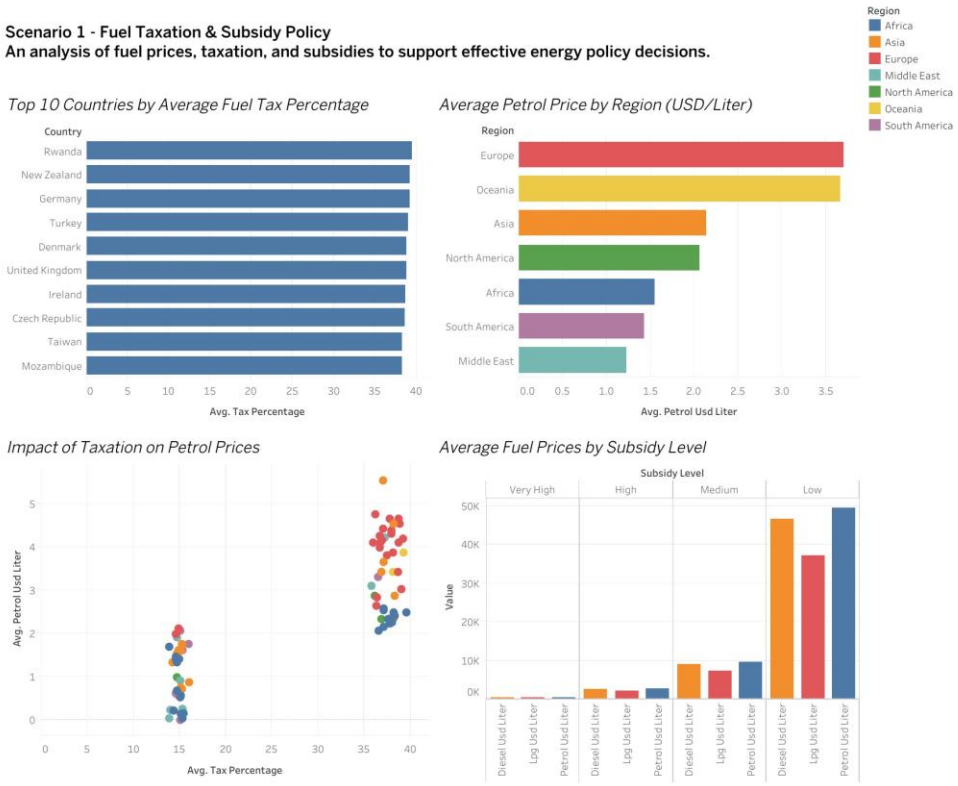
Check area	Evidence-based observation	Status / limitation
Data source	CSV connected directly; dataset notes describe about 27,468 records, 84 countries, 7 regions, Jan 2020–Apr 2026.	Connection and scope documented; no separate ETL pipeline claimed.
Dashboard structure	Three dashboards, four visualisations each.	Visible in supplied screenshots.
Filters and legends	Scenario 3 shows Income Level, Region, and Subsidy Level filters; legends are visible in the relevant dashboards.	Documented from screenshots; no formal usability test claimed.
Calculations	Visuals use Tableau aggregations such as averages and ranking by average petrol price.	No unverified custom calculated fields claimed.
Load/concurrency/mobile	No recorded timing, load, concurrency, or mobile benchmark.	Not formally tested.

7. RESULTS

The primary result is a published Tableau Public workbook with three scenario dashboards. The screenshots below show the latest dashboard images provided for the project. They document the visual structure and visible controls; they are not a substitute for formal usability or performance tests.

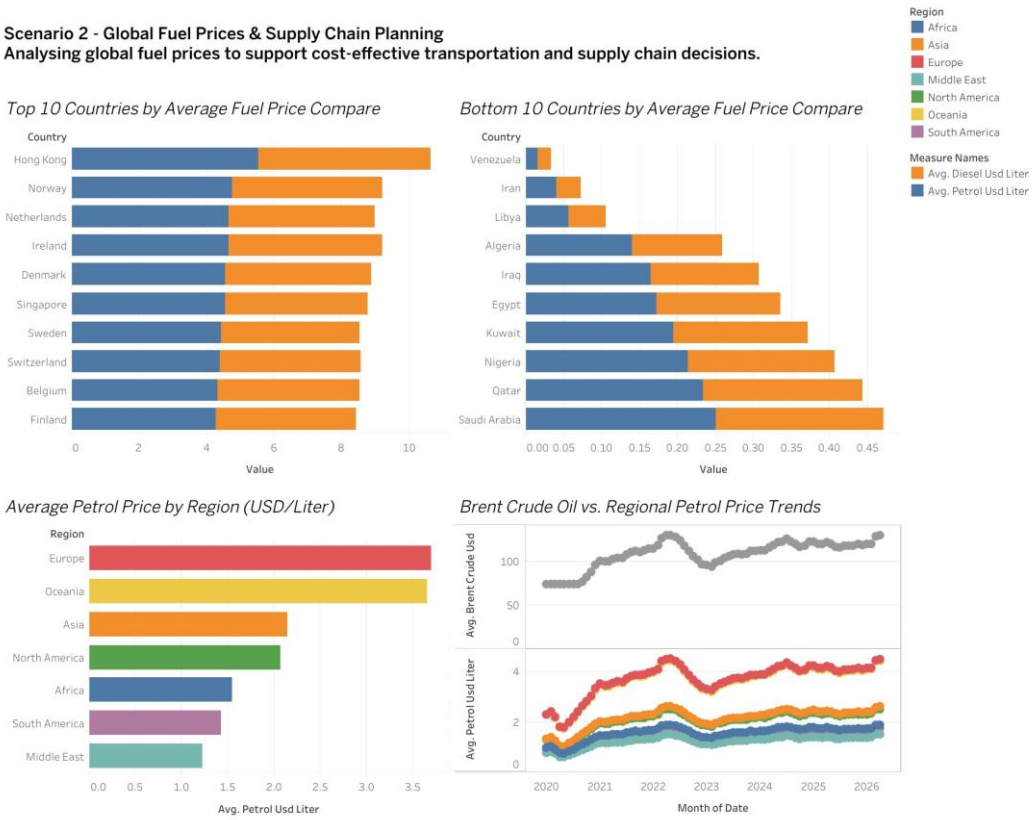
7.1 Output Screenshot — Scenario 1: Fuel Taxation & Subsidy Policy

Shows tax-percentage ranking, regional average petrol prices, tax-versus-petrol-price relationship, and fuel prices by subsidy level.



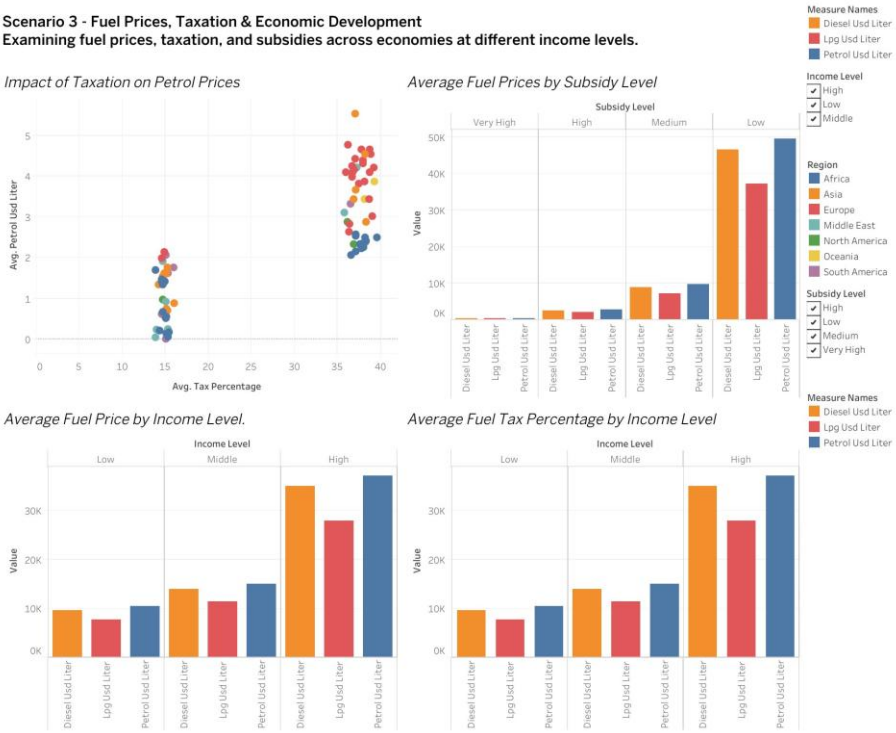
7.2 Output Screenshot — Scenario 2: Global Fuel Prices & Supply Chain Planning

Shows high/low petrol-price country rankings, regional comparison, and historical Brent crude/fuel-price trends.



7.3 Output Screenshot — Scenario 3: Fuel Prices, Taxation & Economic Development

Shows income-level comparisons and visible Income Level, Region, and Subsidy Level filters.



8. ADVANTAGES & DISADVANTAGES

Advantages

- Combines multiple fuel-price and taxation indicators in one published workbook.
- Provides three scenario-focused views instead of requiring users to compare only raw tables.
- Uses country rankings, regional comparisons, category breakdowns, and historical trends to support exploration.
- Visible legends and filters help users interpret colours and narrow some comparisons.
- Tableau Public makes the workbook accessible through a shareable public link.

Disadvantages / Limitations

- The workbook relies on the supplied historical CSV and is not a live fuel-price monitoring feed.
- Averages and group comparisons may hide variation among individual countries or dates.
- Observed associations do not establish causal relationships between taxes, subsidies, income levels, crude oil, and fuel prices.
- The workbook is publicly accessible; it is not suitable for confidential data.
- Formal performance, concurrency, mobile-responsiveness, and usability benchmarks have not been documented.
- No GitHub repository link or public dataset source URL was provided for this report.

9. CONCLUSION

The project demonstrates a practical Tableau workflow: connect a supplied CSV dataset, use built-in aggregations and fields to build analytical worksheets, arrange the visualisations into three dashboards, and publish the workbook on Tableau Public. The resulting views support exploratory comparison of fuel prices, taxation, subsidies, regions, income groups, and historical trends. The project also provides hands-on learning in dashboard design, visual communication, and evidence-based documentation.

The dashboards should be interpreted as descriptive analytical tools. Further validation and additional infrastructure would be needed before presenting the workbook as a live operational system or using its patterns to make causal claims.

10. FUTURE SCOPE

- Create a repeatable data-update and validation procedure for the CSV, then refresh the Tableau workbook when updated data isavailable.
- Evaluate a database or data warehouse only if data volume, update frequency, or governance requirements justify it.
- Conduct structured performance checks for rendering time, filters, workbook responsiveness, and representative user loads.
- Conduct usability checks with representative users and test readability across screen sizes.
- Consider a Tableau Story or guided narrative if required for the final demonstration.
- Explore automated data refresh or integration with a trusted data source only after validating licensing, data quality, and technicalfeasibility.
- If the project requires source control, create a GitHub repository and add the workbook/documentation; no repository URL is currentlyrecorded in this report.

These are future possibilities, not features claimed as implemented. The suggested phases and timelines in the separate Scalability & Future Plan document are planning estimates.

11. APPENDIX

11.1 Source Code (if any)

No separate source-code project was used. The solution was built in Tableau Public using a CSV file, Tableau worksheets, aggregations, dashboard layout, and filters/legends. No Python, SQL, or custom web-application code is claimed.

11.2 Dataset Link

The dataset file used is global_fuel_prices_Dataset.csv, supplied for the project. A public download URL was not provided, so no external dataset link is listed.

11.3 Project Links

The following links provResource	de direct access to the published project resources.Direct Link
Website	https://fuel-prices-taxation-dashboard.onrender.com/
Tableau Public	https://public.tableau.com/app/profile/harshvardhan.gupta1425/viz/Global_Fuel_Prices_and_Taxation_Trends_17932843176370/Story1
Demo Video	https://drive.google.com/file/d/1GLqMmXUegjqNqqAs9VH-QL4tpPlfsRss/view?usp=sharing
GitHub	https://github.com/Harshvardhan-30/global-fuel-prices-taxation-analysis

Scalability & Future Plan

Global Fuel Prices and Taxation Trends: A Comprehensive Global Performance Intelligence Report

Student / Project Type	Platform
Harshvardhan Gupta — Individual Project	Tableau Public 2026.3; CSV dataset

Scalability & Future Plan

Scalability Aspect	Current State	Proposed Upgrade / Solution
User Load	Published on Tableau Public for viewing and exploration; no concurrent-user benchmark recorded.	If usage grows, monitor workbook usage and consider an appropriate managed Tableau deployment or embedded analytics approach, subject to requirements and licensing.
Data Storage	CSV file connected directly to Tableau; no separate database or warehouse used.	For larger or frequently refreshed data, evaluate a managed database or warehouse and define a reliable refresh process. This is future work, not part of the current implementation.

Performance	Three dashboards with four charts each, using the current dataset of approximately 27,468 records.	Profile workbook performance with a documented test plan; optimize filters, aggregations, and visual complexity, and consider extracts or a more suitable data platform if needed.
Security	Workbook is published to Tableau Public, which is intended for publicly accessible content.	Before handling non-public data, review access-control and governance needs and use a platform that supports required permissions. Do not publish confidential data to a public workbook.

This document records the current solution's known limitations and realistic options for future growth. Proposed upgrades are recommendations only; they are not claims about features already implemented.

Current System Limitations

S.No	Current System Limitation	Priority to Address	Impact
1	The workbook uses a static CSV dataset; new records are not automatically sourced from a live external feed.	High	The analysis can become outdated unless the dataset is manually updated and the workbook refreshed.
2	The solution is a Tableau Public workbook with three dashboards, not a custom multi-user web application.	Medium	The experience and access controls depend on Tableau Public and its supported sharing features.
3	Formal load, concurrency, mobile-responsiveness, and performance benchmarks have not been conducted.	Medium	Capacity and response-time limits under heavier use are not quantified.

Scalability Plan

Future Roadmap

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
Phase 1	Maintain and verify the existing three dashboards; review labels, filters, and data quality when the CSV is updated.	Near term (1–2 months; proposed)	Keeps the current analysis understandable and reduces errors from stale or inconsistent source data.
Phase 2	Document a repeatable dataset refresh and validation procedure; evaluate whether a database connection is justified.	Short to medium term (2–4 months; proposed)	Improves repeatability and prepares the project for more frequent or larger data updates.
Phase 3	Conduct structured performance and usability checks, including filter behavior and dashboard readability on different screen sizes.	Medium term (4–6 months; proposed)	Provides evidence about user experience and performance instead of relying on assumptions.

Phase 4	If requirements justify it, investigate automated data refresh, a managed data source, or a Tableau Story/web integration.	Longer term (6+ months; conditional)	Could support more timely insights and a guided narrative, subject to technical feasibility and available tools.

Scope and Accuracy Notes

- Current implementation: a CSV dataset connected directly to Tableau Public and presented through three dashboards; no separate database, automated live feed, or custom web application has beenestablished.
- The proposed timelines are planning estimates, not committed delivery dates.
- The current workbook is public. Do not upload confidential, personal, or restricted data to Tableau Public.
- Any database, automated refresh, Tableau Story, or web integration should be treated as a possible future enhancement and only pursued if required by project expectations.